

RS-DRL Process Diagrams

Thesis Documentation

January 3, 2026

Diagram Map of RS-DRL Process Models

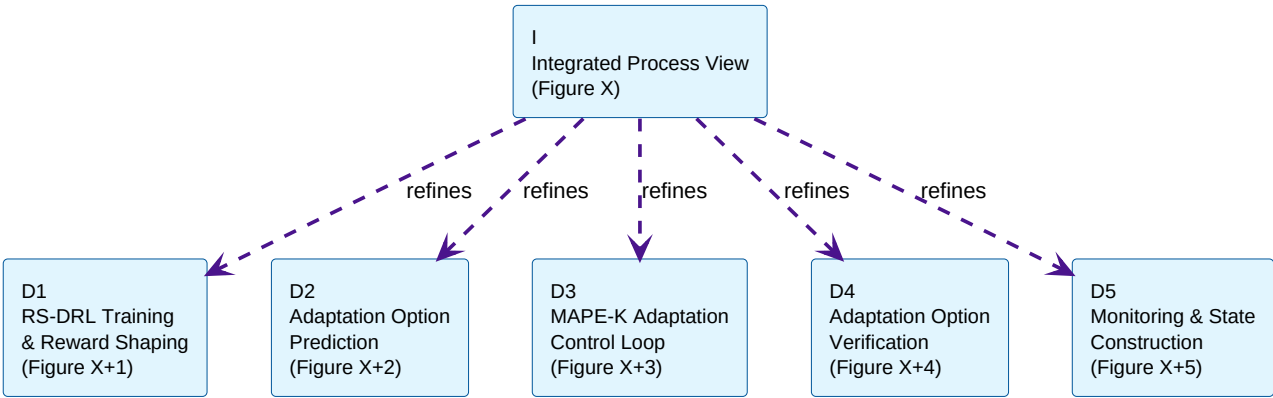


Figure 1: Diagram map of the RS-DRL-based self-adaptive system. The central integration diagram provides a high-level process view, while Diagrams D1–D5 refine the learning, decision-making, verification, control, and monitoring subprocesses presented in this chapter.

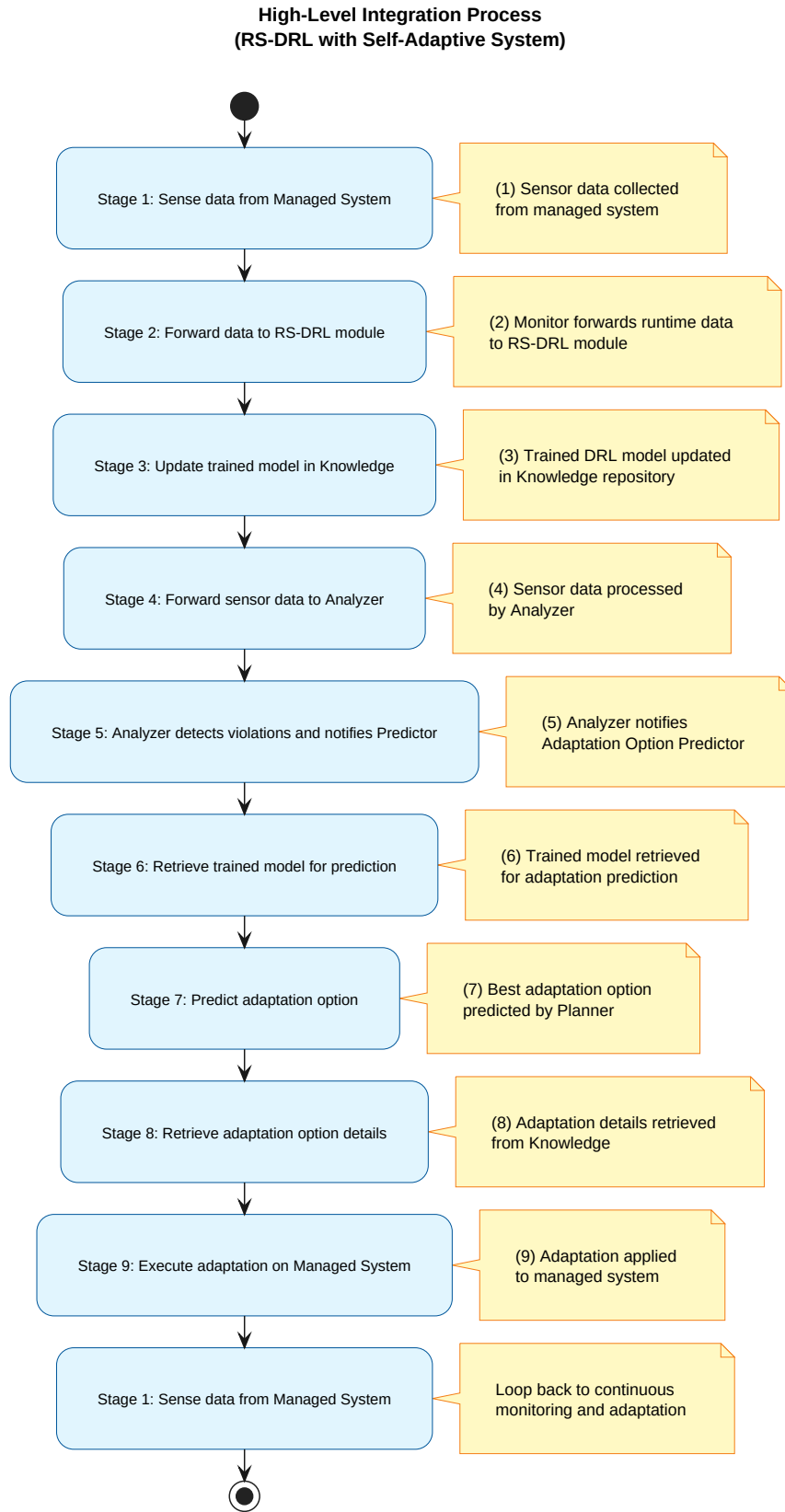


Figure 2: High-level integration process overview showing the nine numbered stages of the RS-DRL-based self-adaptive system. The sequential flow illustrates how data sensing, feature extraction, model updates, analysis, planning, and execution are integrated in a continuous adaptation loop.

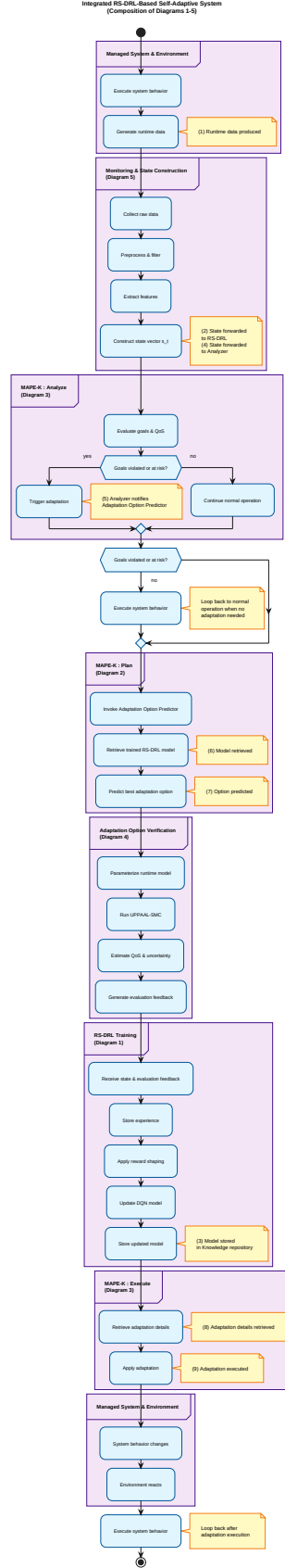


Figure 3: Integrated process view of the RS-DRL-based self-adaptive system. The figure illustrates the closed-loop interaction between monitoring, learning, verification, planning, and execution, and refines the original architectural view presented in Figure 4 of the reference model.

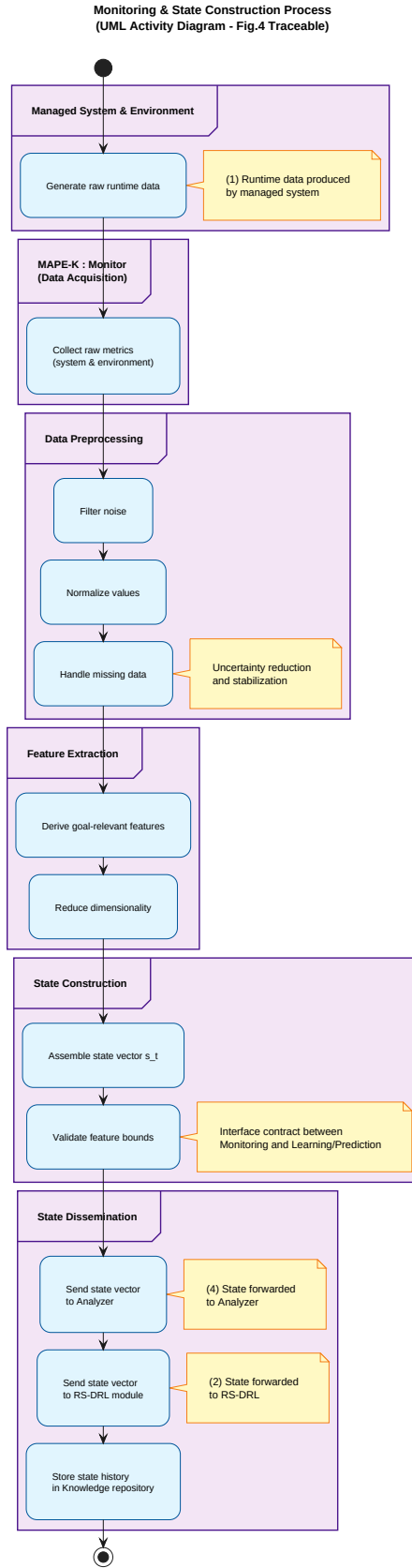


Figure 4: Monitoring and state construction process. Raw runtime data from the managed system and environment is filtered, abstracted, and transformed into a state vector suitable for both RS-DRL learning and runtime adaptation analysis.

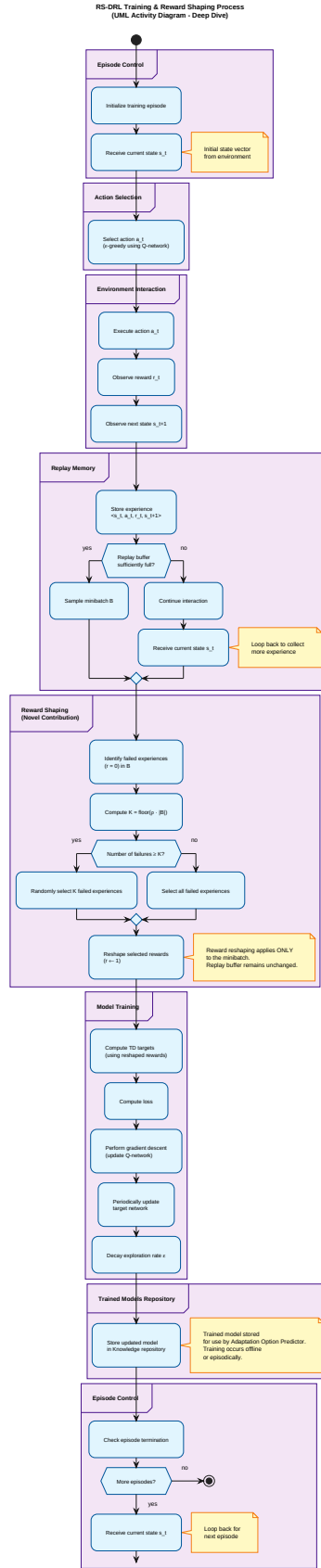


Figure 5: RS-DRL training and reward shaping process. The figure details the episodic interaction with the environment, experience replay, and the novel reward reshaping mechanism applied during minibatch learning.

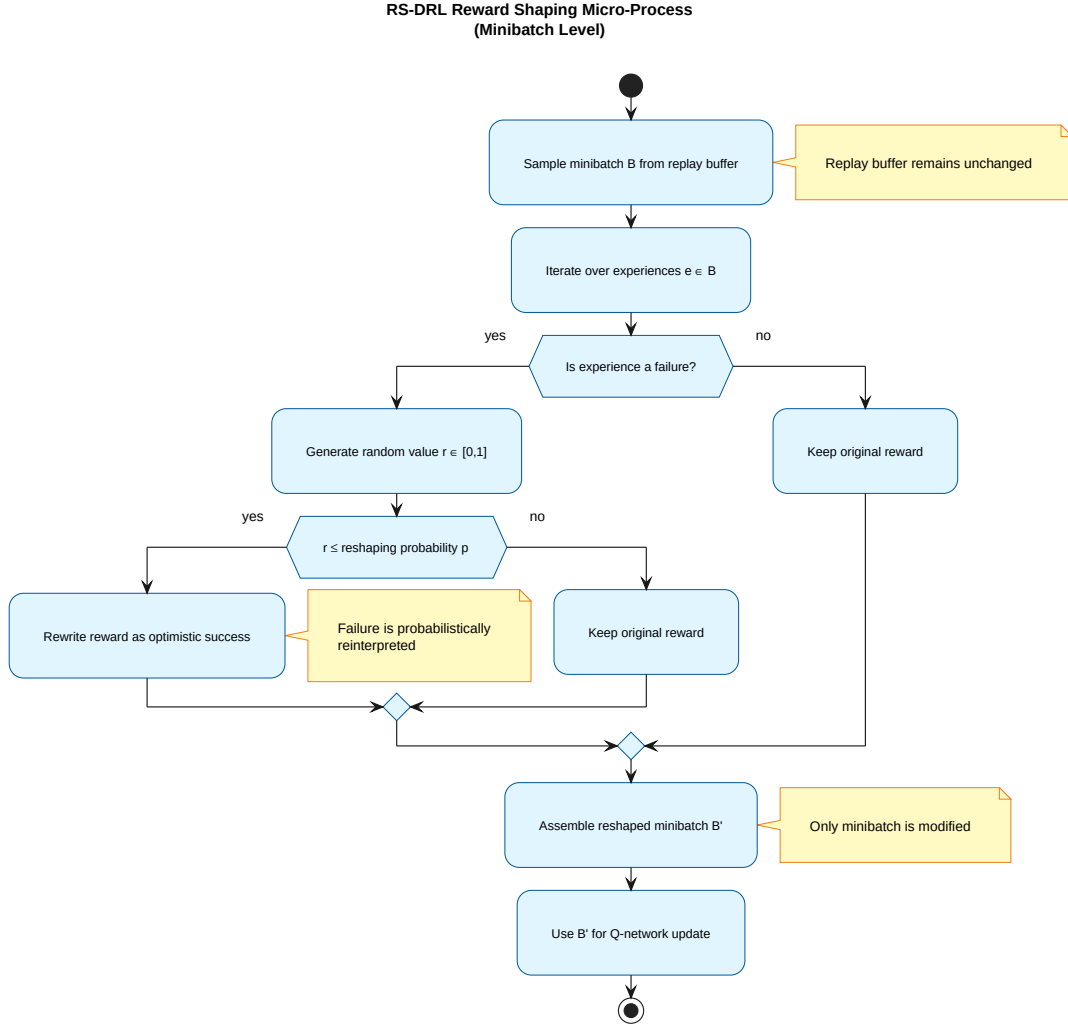


Figure 6: Reward shaping micro-process applied during RS-DRL training. Failed experiences sampled in a minibatch are probabilistically reinterpreted as optimistic successes according to a reshaping probability p . Reward reshaping is applied exclusively to the sampled minibatch and does not modify the replay buffer, preserving experience diversity while accelerating learning convergence.

**Adaptation Option Prediction Process
(Runtime Decision Path - Fig.4 Traceable)**

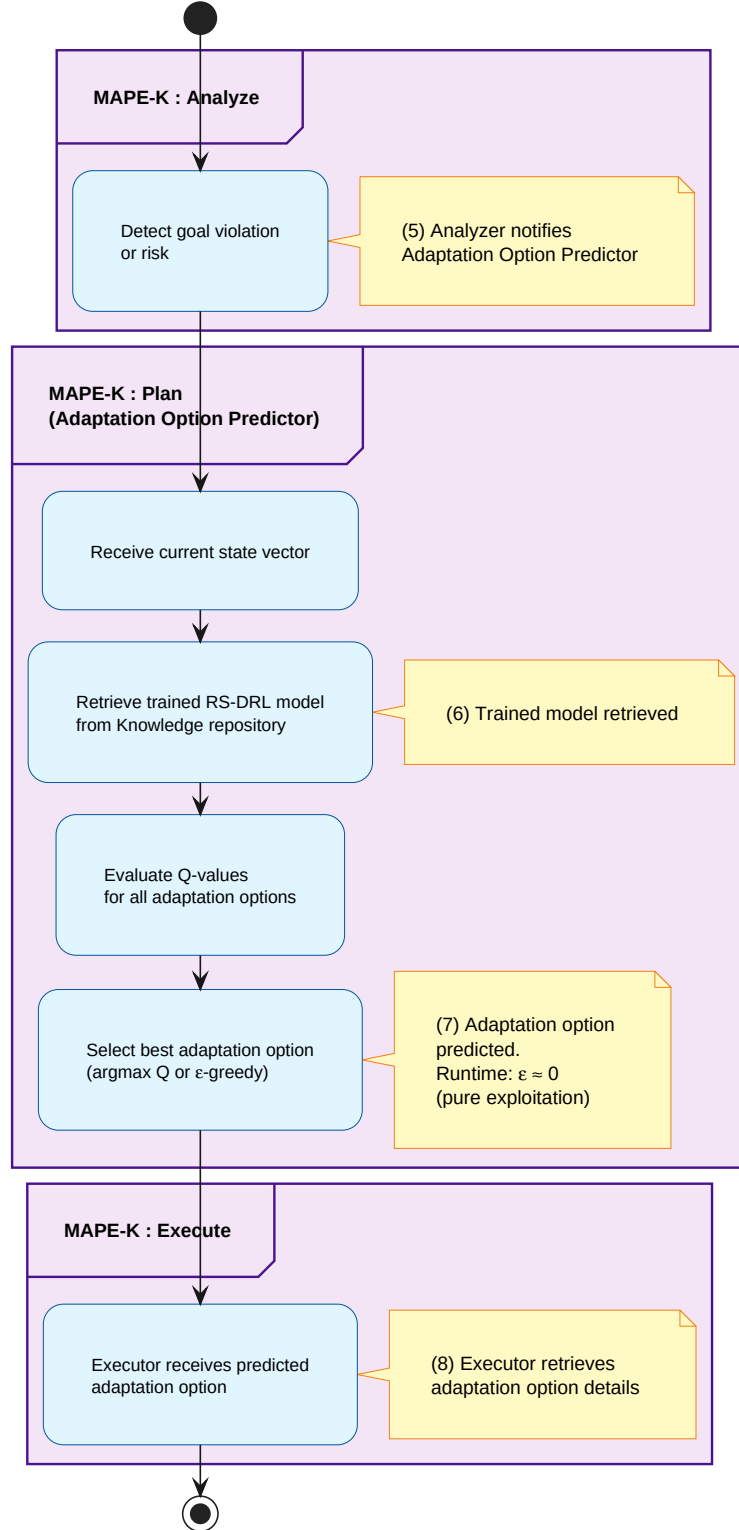


Figure 7: Adaptation option prediction process at runtime. Upon detection of goal violations, the trained RS-DRL model is used to predict the most suitable adaptation option without further learning.

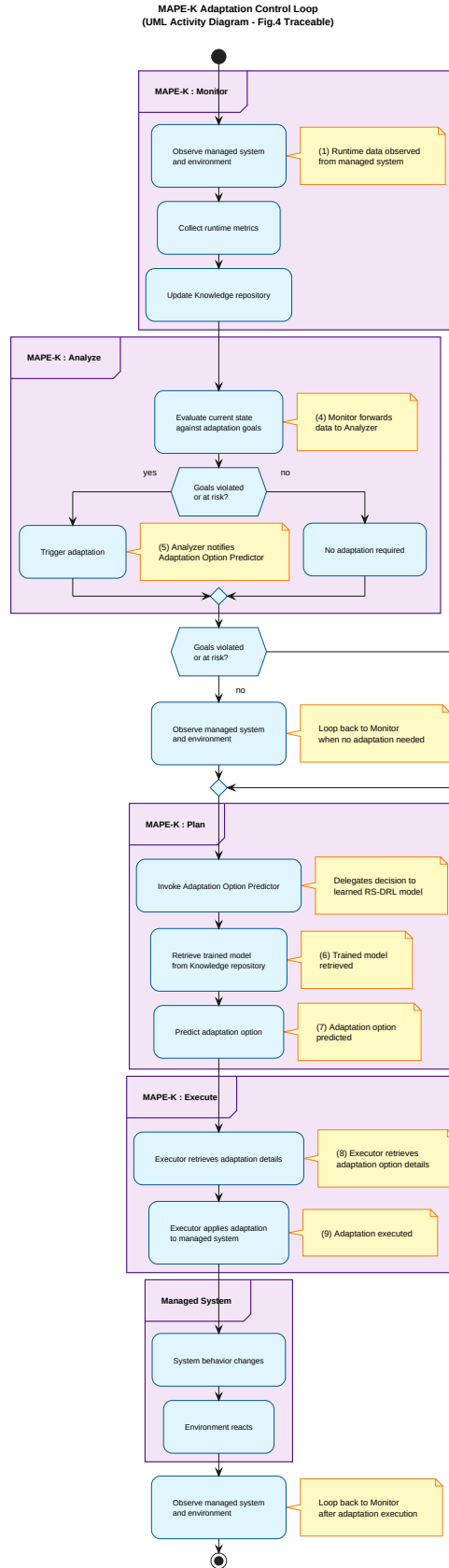


Figure 8: MAPE-K adaptation control loop. The figure illustrates how monitoring and analysis trigger adaptation, how planning delegates decision making to the RS-DRL predictor, and how execution enacts the selected adaptation.

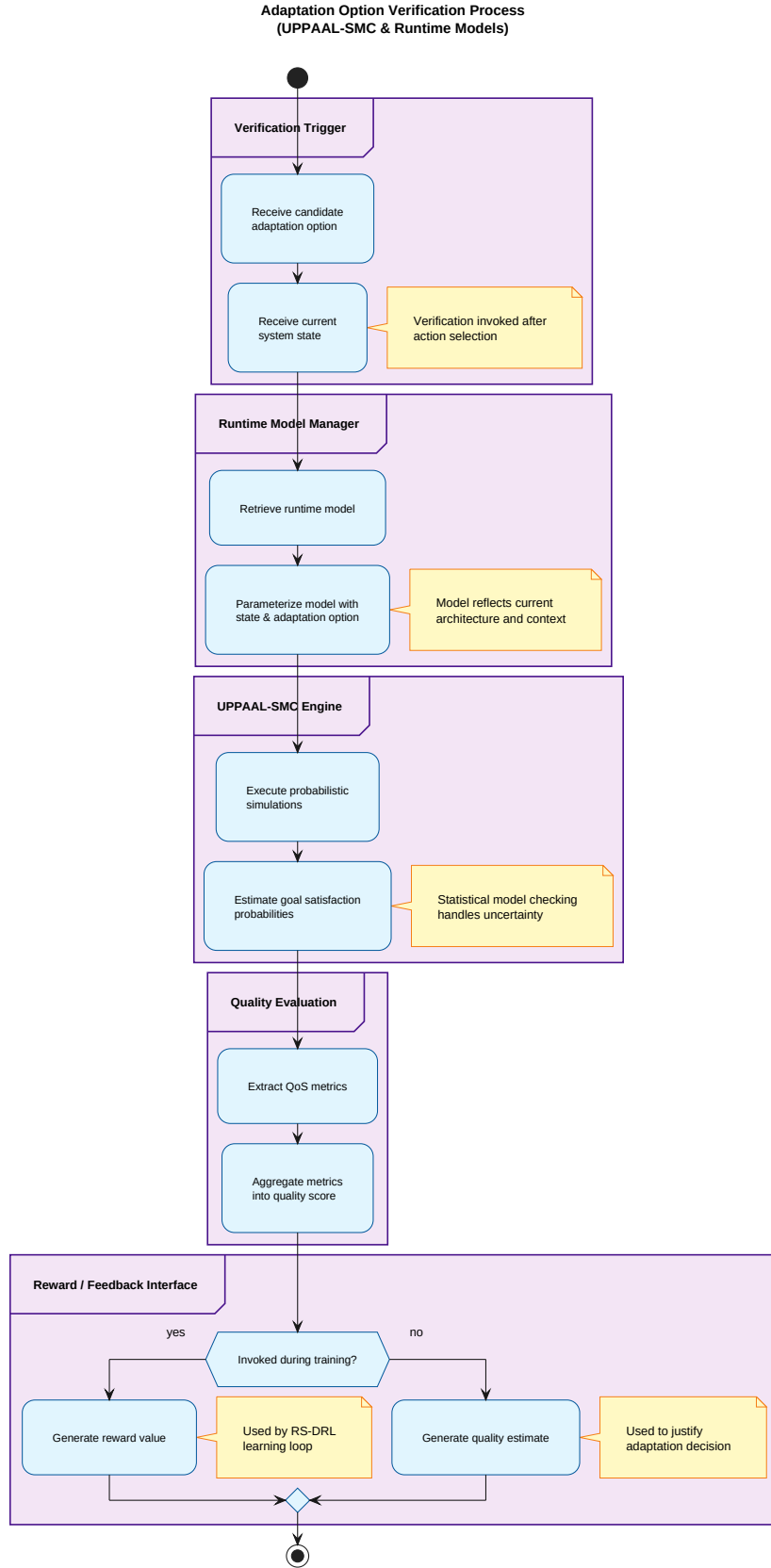


Figure 9: Adaptation option verification using runtime models and statistical model checking. The verification process evaluates the expected quality of adaptation options under uncertainty and provides feedback for learning and decision justification.